Umar Khan

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Experience

Full-Stack Software Engineer — September 2025 - Present
Ingenio Care Inc. — Lombard, IL

- Own 5 provider-facing feature areas end to end, including patient management, clinical workflows, scheduling, billing, and claims, in a multi-tenant healthcare platform handling PHI.
- Designed role-based access control and multi-tenant authorization with JWT, enforcing per-tenant data isolation and API-level permissions across provider organizations.
- Integrated telehealth video, payments, messaging, and AI-assisted clinical workflows via third-party APIs (EHR, Amazon Chime, Stripe, OpenAI).
- Strengthened CI/CD reliability by adding automated health checks and environment validation to GitHub Actions, catching issues before they reached production.

Software Development Intern — January 2025 - September 2025
Ingenio Care Inc. — Lombard, IL

- Built and shipped full-stack features on a healthcare network platform, from database schema through UI.
- Cut bug rates 30% by expanding test coverage and tightening debugging cycles, catching regressions before they reached production.
- Optimized backend services and query performance, improving system responsiveness as the platform's data and user load grew.

Project Experience

Sunnah Lens | Next.js, React, TypeScript, FastAPI, PostgreSQL, pgvector, Gemini, RAG — August 2026

- Built and launched Sunnah Lens (sunnahlens.com), a full-stack AI search web app over 29,653 hadith from the six canonical collections, letting users ask questions in plain English instead of knowing references.
- Developed a RAG chat pipeline that uses an LLM to extract search keywords, embeds the question with Gemini, retrieves the top 8 hadith via pgvector, and streams a grounded, cited answer over Server-Sent Events from FastAPI.
- Engineered prompt guardrails that restrict answers to the retrieved hadith, require inline [n] citations mapped to source cards, and state plainly when the sources do not cover a question, preventing ungrounded responses.
- Implemented semantic search with a pgvector HNSW index and collection and sub-book filters, plus an embedding cache and per-IP rate limiting to control LLM and embedding API costs.

LLM Pipeline | Scala, Apache Spark, AWS EMR, Hadoop MapReduce, DeepLearning4j, JTokkit — November 2024

- Built a distributed LLM training pipeline that processed a large-scale text corpus beyond single-machine memory limits, covering preprocessing, BPE tokenization, and transformer model training.
- Parallelized tokenization, vocabulary construction, and embedding computation across a Hadoop MapReduce and Spark cluster on AWS EMR.
- Developed parameter-sharing and distributed training strategies, tracking loss, gradient norms, and cluster resource utilization to evaluate performance.

Evently | ReactJS, Node.js, Python, Machine Learning — August 2024

- Built Evently, a live React web app (evently.wiki) with personalized event recommendations, growing user engagement 40% over scattered, one-size-fits-all listings.
- Trained a recommendation model on user RSVP history to power personalized event suggestions.
- Engineered a Node.js/Python web-scraping pipeline that aggregates and standardizes real-time event data from multiple external sources.
- Designed and deployed a secure, fraud-resistant QR code check-in system for event attendees.

Smart Grocery | Flutter, Dart, Git, Jira — March 2024

- Built Smart Grocery, a cross-platform Flutter app unifying recipe discovery and grocery-list management, driving a recommendation engine that grew user engagement 35% and recipe conversions 25%.
- Designed a SQL database with hashed-password authentication to manage user profiles and serve 1,000+ recipes.
- Led integration and sprint planning using Git and Jira for Agile task tracking.

Nostalgia Threads | Java, Django, React, Git January 2024

- Built Nostalgia Threads, a full-stack social platform for sharing cultural memories by region and tag, giving communities a dedicated space that generic social apps don't support.
- Implemented a Trie for tag autocomplete and a priority queue to surface trending memories, keeping search and ranking responsive as content volume grew.
- Designed RESTful Django APIs for memory creation, retrieval, tag-based filtering, and real-time reaction updates, paired with a React frontend with a live feed and emoji reactions.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java
Frameworks & Databases: React, Next.js, Node.js, Express, PostgreSQL
Cloud & Tools: AWS, Docker, GitHub Actions, Git
Also: Go, Scala, C/C++, Flutter, Django, Apache Spark, OpenAI API

Education

University of Illinois Chicago, IL
Bachelor of Science in Computer Science May 2025